Historical Multidimensional ModDiv Key Exchange

Corrected Algebra, Reconciliation Limits, and Security Scope

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Historical note. The research branch reconstructed here has been informally called "MODDIV 2020" in later project discussions. The public archival record located for the multidimensional construction is IACR ePrint 2021/701, dated 20 July 2021 [7]. Accordingly, this article does *not* claim public disclosure in 2020. It is a corrected mathematical reconstruction of the historical construction, based only on public references and self-contained proofs.

Abstract

We revisit the historical multidimensional ModDiv key-exchange proposal in which selected coefficients of ordinary polynomial products are reduced modulo an integer and divided coefficientwise. The public observation is a structured Toeplitz/Hankel rounding map, not an independently sampled LWR matrix. We give explicit zero-based index conventions, correct the symmetric cross-product equations, prove the exact central-product identity, and retain the true deterministic rounding remainders. For power-of-two moduli we derive a corrected coarse extraction rule under which the two parties' output coordinates are equal or circularly adjacent. This replaces an ordinary-distance statement that fails at modular wraparound. We then show that circular adjacency alone does not determine the failure probability of any subsequent low-bit deletion: a conditional boundary-crossing law is necessary. The historical identification with "an LWR instance" is therefore treated as structural context rather than as a security reduction. The one-dimensional predecessor is publicly known to admit a practical low-dimensional CVP attack; no analogous general polynomial-time break is proved here for the multidimensional rounded map, but neither is a confidentiality theorem or concrete security level established. Exact arithmetic checks accompany the paper. The result is a corrected research specification and a precise list of what remains unproved, not a deployment recommendation.

Keywords: modular reduction; integer division; Toeplitz matrices; Hankel matrices; polynomial middle products; learning with rounding; key agreement; reconciliation; cryptanalysis.

1 Public record and scope

The scalar ModDiv line was publicly described through the map

$$x \longmapsto \left\lfloor \frac{[ax]_{2^p}}{2^q} \right\rfloor$$

in the public cryptosystem record of Azhari and Bouftass [1]. Zhang subsequently gave a practical attack on the scalar key exchange using a two-dimensional closest-vector formulation and Gaussian lattice reduction [2]. The publisher record for Barcau, Paşol, Pleşca and Togan identifies a later SecITC 2017 study of a key-exchange protocol in this line [3]; the present article uses that reference only as historical context and does not rely on unverified technical claims from that paper.

A multidimensional proposal appears as IACR ePrint 2021/701 [7]. It replaces the scalar multiplication by selected coefficients of an ordinary polynomial product and describes the resulting public observations as a form of learning with rounding. The present paper reconstructs the mathematical core of that proposal while correcting index, wraparound, and probability statements. It does not import any later MODDIV-WC, MODDIV-NOISE, or BLINDAGE mechanism.

The following distinctions are mandatory throughout:

(i) an exact algebraic identity is not a computational hardness theorem;

(ii) a bound on circular distance is not an agreement probability;

(iii) a vector length is not an entropy lower bound or a bit-security estimate;

(iv) resemblance to LWR does not identify the complete joint distribution with a published LWR experiment.

2 Modulo-division and the scalar baseline

For a positive integer M, let $[z]_M \in \{0, \dots, M-1\}$ be the canonical residue of z modulo M.

Definition 2.1 (Modulo-division)**.** *For positive integers P, Q, define*

$$\mathrm{MD}_{P,Q}(z) = \left\lfloor \frac{[z]_P}{Q} \right\rfloor.$$

When $P = 2^p$ and $Q = 2^q$, we may write $\mathrm{MD}^{(2)}_{p,q}$.

Modulo-division is not additive because the discarded remainder generates carries. This point matters already in the scalar key exchange.

Assume for the moment that p, q are integers with $q \geq 1$ and $p > 2q$, set $P = 2^p$ and $Q = 2^q$, and let

$$U = \mathrm{MD}_{P,Q}(AX), \qquad V = \mathrm{MD}_{P,Q}(AY).$$

Consider the symmetric coarse values

$$W_A = \left\lfloor \frac{[QXV]_P}{Q^2} \right\rfloor, \qquad W_B = \left\lfloor \frac{[QYU]_P}{Q^2} \right\rfloor.$$

Let $K = P/Q^2$ and define circular distance on $\mathbb{Z}/K\mathbb{Z}$ by

$$\mathrm{dist}_K(a, b) = \min\{[a-b]_K, [b-a]_K\}.$$

Theorem 2.2 (Scalar circular closeness)**.** *Let $T = [AXY]_P$ and $C = \lfloor T/Q^2 \rfloor$. If $0 \leq X, Y < Q$, then*

$$W_A, W_B \in \{C, [C-1]_K\}, \qquad \mathrm{dist}_K(W_A, W_B) \leq 1.$$

Proof. Write

$$[AY]_P = QV + e_Y, \qquad [AX]_P = QU + e_X,$$

with $0 \leq e_X, e_Y < Q$. Multiplying by the opposite secret gives

$$QXV \equiv AXY - Xe_Y \pmod{P}, \qquad QYU \equiv AXY - Ye_X \pmod{P}.$$

Because $0 \leq Xe_Y, Ye_X \leq (Q-1)^2 < Q^2$, subtracting either error from the same residue T can lower the quotient by at most one coarse cell. At the modular origin the value -1 is represented as $K-1$, hence the conclusion is circular rather than an ordinary absolute-distance statement. □

Example 2.3 (Wraparound)**.** *Take $P = 64$, $Q = 4$, $A = 43$, $X = 2$, $Y = 3$. Then $U = 5$, $V = 0$, and*

$$W_A = 0, \qquad W_B = 3 \quad \textit{in } \mathbb{Z}/4\mathbb{Z}.$$

Thus $|W_A - W_B| = 3$ but $\mathrm{dist}_4(W_A, W_B) = 1$.

This example is important for the multidimensional reconstruction: ordinary integer distance is not the correct invariant after modular wraparound.

3 The historical central polynomial map

3.1 Extraction operator

For a polynomial $Z(x) = \sum_j z_j x^j$ and integers $0 \le a < b$, define

$$\Pi_{b,a}(Z) = \sum_{i=0}^{b-a-1} z_{a+i} x^i.$$

Equivalently, $\Pi_{b,a}$ retains coefficients of degrees $a, \dots, b-1$ and shifts them down by a. This is ordinary truncation in $\mathbb{Z}[x]$; no cyclic or negacyclic wraparound is used.

Fix integers $n \ge 0$ and $m > 2n$. Write

$$A(x) = \sum_{i=0}^{m-1} a_i x^i, \quad X(x) = \sum_{j=0}^{n} x_j x^j, \quad Y(x) = \sum_{j=0}^{n} y_j x^j.$$

The secret vectors therefore have

$$N = n + 1$$

coefficients. The first extracted vectors have length

$$L = m - n,$$

and the final central vectors have length

$$h = m - 2n.$$

For the power-of-two reconstruction let p, q be integers with $q \ge 1$ and set

$$P = 2^p, \qquad Q = 2^q.$$

with $a_i \in \{0, \dots, P-1\}$ and $x_j, y_j \in \{0, \dots, Q-1\}$. These full ranges are used to state deterministic identities cleanly. This is a deliberately fixed power-of-two reconstruction. In the public 2021 presentation, p and q are used as bit-size parameters for general integers P and Q, rather than as identities $P = 2^p$ and $Q = 2^q$ throughout, and its reconciliation parameter is described locally rather than as the single public integer fixed below. The present choices of power-of-two moduli, $N = n + 1$ secret coefficients, corrected window widths, and a fixed public coarse parameter are therefore part of the reconstructed model, not claims of literal identity with every historical sampling rule. A sampler requiring coefficients of exact bit length is a subdomain of this fixed deterministic model, but it induces a different probability law; no agreement or hardness probability is transferred between the two samplers without proof.

Define the raw central products

$$U = \Pi_{m,n}(AX), \qquad V = \Pi_{m,n}(AY),$$

and their coefficientwise public quotients

$$U_1 = \mathrm{MD}_{P,Q}(U), \qquad V_1 = \mathrm{MD}_{P,Q}(V).$$

Here MD acts coefficientwise.

The private cross-products are reconstructed symmetrically as

$$C_A = \Pi_{m-n,n}(XV_1), \qquad C_B = \Pi_{m-n,n}(YU_1).$$

This is the zero-based form of the shifted extraction printed historically: explicitly,

$$\Pi_{m-n,n}(XV_1) = \Pi_{m,2n}(x^n XV_1), \qquad \Pi_{m-n,n}(YU_1) = \Pi_{m,2n}(x^n YU_1).$$

The use of YU_1 in Bob's equation is required by the intended symmetry. The public 2021 text contains a line with XU_1; the central-product identity below shows why that line cannot be the symmetric counterpart of Alice's computation.

Each public raw coordinate is

$$U_i = \sum_{j=0}^{n} a_{n+i-j} x_j, \qquad 0 \le i < L.$$

Hence the public linear operator is Toeplitz. Reversing secret coordinates gives the equivalent Hankel representation. Adjacent rows overlap heavily in the public coefficients.

3.2 Exact central-product identity

Lemma 3.1 (Central-product identity)**.** *Under the dimensions above,*

$$\Pi_{m-n,n}(X\,\Pi_{m,n}(AY)) = \Pi_{m,2n}(AXY) = \Pi_{m-n,n}(Y\,\Pi_{m,n}(AX))\,.$$

Proof. For $0 \le i < h$, the coefficient on the left is

$$\sum_{j=0}^{n} x_j (AY)_{2n+i-j} = (AXY)_{2n+i}.$$

The indices $2n + i - j$ lie in $[n, m-1]$, so every required coefficient survives the inner extraction $\Pi_{m,n}$. The second equality follows by interchanging X and Y and using commutativity of ordinary polynomial multiplication. □

The identity depends on the index range, not on commutativity alone. It would need to be rederived for cyclic or negacyclic products.

4 Exact remainders and corrected coarse outputs

Let E_X, E_Y be the true coefficientwise remainders of the raw public products:

$$(E_X)_i = [U_i]_P - Q(U_1)_i, \qquad (E_Y)_i = [V_i]_P - Q(V_1)_i.$$

Thus

$$0 \le (E_X)_i, (E_Y)_i < Q.$$

Put

$$T = \Pi_{m,2n}(AXY),$$
$$R_A = \Pi_{m-n,n}(XE_Y), \qquad R_B = \Pi_{m-n,n}(YE_X).$$

Lemma 4.1 (Exact remainder identities)**.** *Coefficientwise modulo P,*

$$QC_A \equiv T - R_A \pmod{P}, \qquad QC_B \equiv T - R_B \pmod{P}.$$

Moreover every coordinate satisfies

$$0 \le (R_A)_i, (R_B)_i \le B := N(Q-1)^2.$$

Proof. The quotient decompositions give $QV_1 \equiv V - E_Y$ and $QU_1 \equiv U - E_X$ modulo P. Apply $\Pi_{m-n,n}$ after multiplying by X or Y, then use the central-product identity. Each error coordinate is a sum of N nonnegative products, each at most $(Q-1)^2$. □

Set

$$\ell = \lceil \log_2 N \rceil,$$

fix a public integer $r \geq 0$ before the exchange, and assume

$$p > 2q + \ell + r.$$

Define

$$D = 2^{2q+\ell+r}, \qquad K = P/D = 2^{p-2q-\ell-r}.$$

For each retained coordinate define

$$(K_A)_i = \left\lfloor \frac{[Q(C_A)_i]_P}{D} \right\rfloor, \qquad (K_B)_i = \left\lfloor \frac{[Q(C_B)_i]_P}{D} \right\rfloor.$$

Theorem 4.2 (Polynomial circular closeness)**.** *For every admissible coefficient choice and every retained coordinate i, let*

$$c_i = \left\lfloor \frac{[T_i]_P}{D} \right\rfloor.$$

Then

$$(K_A)_i, (K_B)_i \in \{c_i, [c_i - 1]_K\},$$

and therefore

$$\mathrm{dist}_K((K_A)_i, (K_B)_i) \leq 1.$$

Proof. Since $2^\ell \geq N$ and $Q \geq 2$,

$$B = N(Q-1)^2 < NQ^2 \leq 2^\ell Q^2 \leq D.$$

The exact remainder identities therefore express both $Q(C_A)_i$ and $Q(C_B)_i$ as the same target T_i minus an integer in $[0, D)$. The scalar quotient argument applies with coarse-cell width D, including modular wraparound. □

The reconstructed output vector has $h = m - 2n$ coordinates and representation length

$$h \log_2 K = (m - 2n)(p - 2q - \ell - r)$$

bits. This is only a representation size. It is not a lower bound on conditional entropy and not a security level.

5 Why reconciliation needs a joint law

The theorem above gives only equality or circular adjacency coordinatewise. It does not determine the probability that further deletion of low bits makes two coordinates equal.

Let $W_A, W_B \in \mathbb{Z}/K\mathbb{Z}$ be equal or adjacent, and for an integer $s \geq 0$ define the canonical coarsening

$$G_s(w) = \left\lfloor \frac{w}{2^s} \right\rfloor.$$

Let

$$\mathcal{D} = \{W_A \neq W_B\}$$

and let $\mathcal{B}_s$ be the event that an adjacent unequal pair lies in two *distinct* G_s blocks; this includes a crossing at the circular origin whenever the two canonical representatives receive different G_s labels. Whenever $\Pr(\mathcal{D}) > 0$,

$$\Pr[G_s(W_A) \neq G_s(W_B)] = \Pr(\mathcal{D}) \Pr(\mathcal{B}_s \mid \mathcal{D}).$$

If $\Pr(\mathcal{D}) = 0$, the left-hand side is zero and no conditional probability is needed. If $2^s \geq K$, then G_s is constant on the canonical alphabet $\{0, \ldots, K-1\}$, so $\mathcal{B}_s$ is empty and the coarsened disagreement probability is zero. Thus replacing the conditional factor by 2^{-s} in the nontrivial regime is an additional distributional assertion, not an algebraic consequence.

Example 5.1 (Uniform marginals do not determine reconciliation)**.** *Let $K = 8$, $s = 1$, and let W_A be uniform.*

If $W_B = W_A \oplus 1$, both marginals are uniform, every pair is adjacent, and

$$G_1(W_A) = G_1(W_B)$$

always.

Instead define

$$W_B = \begin{cases} [W_A - 1]_8, & W_A \text{ even}, \\ [W_A + 1]_8, & W_A \text{ odd}. \end{cases}$$

The marginals remain uniform and every pair remains adjacent, but now

$$G_1(W_A) \neq G_1(W_B)$$

always.

As an exact finite scalar illustration, take $P = 64$, $Q = 4$, A uniform on $\{32, \dots, 63\}$, and X, Y independently uniform on $\{2, 3\}$. Direct enumeration gives agreement $15/16$ before the extra division and $31/32$ after division by two. This finite computation is not a universal agreement law.

For a vector of h coordinates, componentwise adjacency also does not imply only three global candidates. Naive coordinatewise reconciliation can produce exponentially many combinations unless additional structure is proved.

6 General moduli and exact output alphabets

The map $\mathrm{MD}_{P,Q}$ is defined for every pair of positive integers. If

$$P = aQ + r_0, \qquad 0 \le r_0 < Q,$$

then the output alphabet is

$$\{0, \dots, \lceil P/Q \rceil - 1\}.$$

For a uniform residue modulo P, all full quotient cells have mass Q/P and the final incomplete cell, when $r_0 > 0$, has mass r_0/P. Hence a uniform residue does not in general produce a uniform quotient when $Q \nmid P$.

The exact polynomial remainder identities remain meaningful for general P, Q, but the power-of-two rescaling used in the coarse extraction above must not be transported unchanged. Divisibility, output-cell geometry, and the chosen final divisor belong to the theorem statement.

7 Security reassessment

7.1 The scalar predecessor is not a hardness foundation

The one-dimensional predecessor is known to admit an efficient practical attack in the parameter regime studied by Zhang, who reduces the inverse relation to a two-dimensional CVP instance and applies lattice reduction [2]. Therefore the historical scalar inversion assumption cannot serve as an unqualified security foundation for the multidimensional construction.

This does not automatically break the multidimensional central-window map: the public transcript and its dimension are different. Conversely, the absence of a transferred scalar attack is not evidence of multidimensional hardness.

7.2 Structured rounding is not automatically standard LWR

Classical LWR was introduced and analyzed for precisely specified public distributions and rounding maps [4, 5]. The present public observation has the form

$$(U_1)_i = \left\lfloor \frac{\left[\sum_{j=0}^{n} a_{n+i-j} x_j\right]_P}{Q} \right\rfloor, \qquad 0 \le i < L,$$

where neighboring rows reuse coefficients of one public Toeplitz ribbon and all rows share the same short secret.

The public operator is therefore structured. A hardness transfer would need an explicit reduction preserving at least:

- the public-ribbon law and all row correlations;
- the secret law and dimension;
- the number of observations;
- the exact quotient map and any incomplete cells;
- the complete key-agreement transcript and reconciliation leakage;
- the adversarial target: original-secret recovery, compatible-preimage search, or key prediction.

Middle-product LWR provides a closer published point of comparison, but its theorems also depend on exact roles, lengths, distributions, rank conditions, and a specific cryptographic construction [6]. Algebraic resemblance is not a reduction.

7.3 Security status

The corrected reconstruction establishes:

(1) an exact central-product identity;

(2) exact deterministic remainder equations;

(3) a sufficient coordinatewise circular-closeness theorem under explicit power-of-two conditions;

(4) exact counterexamples showing that marginal uniformity and adjacency do not determine reconciliation probability.

It does *not* establish:

(1) a universal reconciliation-failure formula;

(2) a reduction from standard LWE/LWR or MP-LWR to this exact public distribution;

(3) IND-CPA, IND-CCA, authenticated key exchange, or KEM security;

(4) a concrete 128-bit security level;

(5) a universal efficient attack on the multidimensional rounded map.

These are deliberately separate statements.

8 Reproducibility

A companion script, `verify_a0_v1_1.py`, uses exact integer arithmetic and writes its JSON output to a caller-selected path or, by default, beside the script. In the frozen run used for this article it verified:

- the scalar wraparound witness $W_A = 0, W_B = 3$ in $\mathbb{Z}/4\mathbb{Z}$;
- the exact toy agreement fractions 15/16 and 31/32;
- 20,000 deterministic-seed polynomial instances of the central-product identity;
- 20,000 instances of both exact remainder identities;
- 20,000 instances of the circular-closeness theorem;
- two eight-state reconciliation countermodels with identical uniform marginals and opposite coarsening behavior.

The tests are consistency checks. The theorems are justified by the proofs above, not by random testing. The code does not attempt to certify computational hardness.

9 Conclusion

The historical multidimensional ModDiv proposal contains a valid and useful algebraic mechanism: central polynomial windows commute in the precise index range required by the protocol, and deterministic quotient remainders can be tracked exactly. Once the remainder terms and modular wraparound are retained, the strongest unconditional correctness statement obtained here is coordinatewise circular closeness under explicit size and divisibility conditions.

That result is weaker than a negligible disagreement theorem and logically independent of confidentiality. A secure descendant would need a specified joint distribution, a complete reconciliation rule, a reduction from a justified source problem, explicit loss accounting, and cryptanalysis of the actual structured transcript. The public scalar attack and the published LWR literature are essential context, but neither may be used as a shortcut in the wrong direction.

Accordingly, this paper preserves the historical ModDiv mechanism while replacing ambiguous security language by explicit mathematical obligations. It is a corrected research baseline for subsequent work, not a deployment profile.

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